1. The voltage gain of an amplifier without feedback is 3000. Calculate the voltage gain of the amplifier if negative voltage feedback is introduced in the circuit. Given that feedback fraction mv = 0.01.
   Solution,
   ∴ Voltage gain with negative feedback is

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v} = \frac{3000}{1 + 3000 \times 0.01} = \frac{3000}{31} = 97$$

2. The overall gain of a multistage amplifier is 140. When negative voltage feedback is applied, the gain is reduced to 17.5. Find the fraction of the output that is fedback to the input.

Solution,

Let m_v be the feedback fraction. Voltage gain with negative feedback is

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$17.5 = \frac{140}{1 + 140\, m_v}$$

or $$17.5 + 2450\, m_v = 140$$

∴ $$m_v = \frac{140 - 17.5}{2450} = \frac{1}{20}$$

3. When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50.
   (i) Calculate the fraction of the output voltage fedback.
   (ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.

   Solution,

(*i*) Gain without feedback, $A_v = 100$

Gain with feedback, $A_{vf} = 50$

Let m_v be the fraction of the output voltage fedback.

Now $$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$50 = \frac{100}{1 + 100\, m_v}$$

or $$50 + 5000\, m_v = 100$$

or $$m_v = \frac{100 - 50}{5000} = \mathbf{0.01}$$

(*ii*) $$A_{vf} = 75\,; \quad m_v = 0.01\,; \quad A_v = ?$$

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$75 = \frac{A_v}{1 + 0.01\, A_v}$$

or $$75 + 0.75\, A_v = A_v$$

$\therefore$ $$A_v = \frac{75}{1 - 0.75} = \mathbf{300}$$

4. With a negative voltage feedback, an amplifier gives an output of 10 V with an input of 0.5 V. When feedback is removed, it requires 0.25 V input for the same output. Calculate (i) gain without feedback (ii) feedback fraction mv .

Solution,

(*i*) Gain without feedback, $A_v = 10/0.25 = \mathbf{40}$

(*ii*) Gain with feedback, $A_{vf} = 10/0.5 = 20$

Now $$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$20 = \frac{40}{1 + 40\, m_v}$$

or $$20 + 800\, m_v = 40$$

or $$m_v = \frac{40 - 20}{800} = \mathbf{\frac{1}{40}}$$

5. The gain of an amplifier without feedback is 50 whereas with negative voltage feedback, it falls to 25. If due to ageing, the amplifier gain falls to 40, find the percentage reduction in stage gain (i) without feedback and (ii) with negative feedback.
Solution,

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$25 = \frac{50}{1 + 50\, m_v}$$

or $$m_v = 1/50$$

(*i*) **Without feedback.** The gain of the amplifier without feedback is 50. However, due to ageing, it falls to 40.

$$\therefore \quad \text{\%age reduction in stage gain} = \frac{50 - 40}{50} \times 100 = \mathbf{20\%}$$

(*ii*) **With negative feedback.** When the gain without feedback was 50, the gain with negative feedback was 25. Now the gain without feedback falls to 40.

$$\therefore \quad \text{New gain with negative feedback} = \frac{A_v}{1 + A_v\, m_v} = \frac{40}{1 + (40 \times 1/50)} = 22.2$$

$$\therefore \quad \text{\%age reduction in stage gain} = \frac{25 - 22.2}{25} \times 100 = \mathbf{11.2\%}$$

6. An amplifier has a voltage amplification Av and a fraction mv of its output is fedback in opposition to the input. If mv = 0.1 and Av = 100, calculate the percentage change in the gain of the system if Av falls 6 db due to ageing.
Solution,

$$A_v = 100, \quad m_v = 0.1, \quad A_{vf} = ?$$

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v} = \frac{100}{1 + 100 \times 0.1} = 9.09$$

$$\text{Fall in gain} = 6db$$

Let A_{v1} be the new absolute voltage gain without feedback.

Then, $\quad 20 \log_{10} A_v/A_{v1} = 6$

or $\quad \log_{10} A_v/A_{v1} = 6/20 = 0.3$

or $\quad \dfrac{A_v}{A_{v1}} = \text{Antilog } 0.3 = 2$

or $\quad A_{v1} = A_v/2 = 100/2 = 50$

$\therefore \quad$ New $\quad A_{vf} = \dfrac{A_{v1}}{1 + A_{v1}\, m_v} = \dfrac{50}{1 + 50 \times 0.1} = 8.33$

$$\text{\% age change in system gain} = \frac{9.09 - 8.33}{9.09} \times 100 = \mathbf{8.36\%}$$

7. An amplifier has a voltage gain of 500 without feedback. If a negative feedback is applied, the gain is reduced to 100. Calculate the fraction of the output fed back. If, due to ageing of components, the gain without feedback falls by 20%, calculate the percentage fall in gain with feedback.

Solution,

$$A_v = 500 \,;\, A_{vf} = 100 \,;\, m_v = ?$$

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $\quad 100 = \dfrac{500}{1 + 500\, m_v}$

$\therefore \quad m_v = \mathbf{0.008}$

Now $\quad A_v = \dfrac{80}{100} \times 500 = 400 \,;\, m_v = 0.008 \,;\, A_{vf} = ?$

$$A_{vf} = \frac{A_v}{1 + A_v\, m_v} = \frac{400}{1 + 400 \times 0.008} = \frac{400}{4.2} = 95.3$$

$\therefore \quad$ % age fall in $A_{vf} = \dfrac{100 - 95.3}{100} \times 100 = \mathbf{4.7\%}$

Note that without negative feedback, the change in gain is 20%. However, when negative feedback is applied, the change in gain (4.7%) is much less. This shows that negative feedback provides voltage gain stability.

8. An amplifier has an open-loop gain Av = 100,000. A negative feedback of 10 db is applied. Find (i) voltage gain with feedback (ii) value of feedback fraction mv .

Solution,

(*i*) *db* voltage gain without feedback

$$= 20 \log_{10} 100{,}000 = 20 \log_{10} 10^5 = 100\ db$$

Voltage gain with feedback $= 100 - 10 = 90\ db$

Now $\quad 20 \log_{10} (A_{vf}) = 90$

or $\quad \log_{10} (A_{vf}) = 90/20 = 4.5$

$\therefore \quad A_{vf} = \text{Antilog } 4.5 = \mathbf{31622}$

(*ii*) $$A_{vf} = \frac{A_v}{1 + A_v\, m_v}$$

or $$31622 = \frac{100{,}000}{1 + 100{,}000 \times m_v}$$

$\therefore \quad m_v = \mathbf{2.17 \times 10^{-5}}$

9. An amplifier is required with a voltage gain of 100 which does not vary by more than 1%. If it is to use negative feedback with a basic amplifier the voltage gain of which can vary by 20%, determine the minimum voltage gain required and the feedback factor.

Solution,

$$100 = \frac{A_v}{1 + A_v\, m_v}$$

or $$100 + 100\, A_v\, m_v = A_v \qquad \ldots (i)$$

Also $$99 = \frac{0.8\, A_v}{1 + 0.8\, A_v\, m_v}$$

or $$99 + 79.2\, A_v\, m_v = 0.8\, A_v \qquad \ldots (ii)$$

Multiplying eq (*i*) by 0.792, we have,

$$79.2 + 79.2\, A_v\, m_v = 0.792\, A_v \qquad \ldots (iii)$$

Subtracting [(*ii*) – (*iii*)], we have,

$$19.8 = 0.008\, A_v \quad \therefore \quad A_v = \frac{19.8}{0.008} = \mathbf{2475}$$

Putting the value of A_v (= 2475) in eq. (*i*), we have,

$$100 + 100 \times 2475 \times m_v = 2475$$

$$\therefore \quad m_v = \frac{2475 - 100}{100 \times 2475} = \mathbf{0.0096}$$

10. The gain and distortion of an amplifier are 150 and 5% respectively without feedback. If the stage has 10% of its output voltage applied as negative feedback, find the distortion of the amplifier with feedback.

Solution,

Gain without feedback, $A_v = 150$

Distortion without feedback, D = 5% = 0.05

Feedback fraction, m_v = 10% = 0.1

If D_{vf} is the distortion with negative feedback, then,

$$D_{vf} = \frac{D}{1 + A_v\ m_v} = \frac{0.05}{1 + 150 \times 0.1} = 0.00313 = \mathbf{0.313\%}$$

It may be seen that by the application of negative voltage feedback, the amplifier distortion is reduced from 5% to 0.313%.

11. An amplifier has a gain of 1000 without feedback and cut-off frequencies are f1 = 1.5 kHz and f2 = 501.5 kHz. If 1% of output voltage of the amplifier is applied as negative feedback, what are the new cut-off frequencies?

Solution,

$$A_v = 1000\ ; \quad m_v = 0.01$$

The new lower cut-off frequency with feedback is

$$f_{1(f)} = \frac{f_1}{1 + A_v\ m_v} = \frac{1.5\ \text{kHz}}{1 + 1000 \times 0.01} = \mathbf{136.4\ Hz}$$

The new upper cut-off frequency with feedback is

$$f_{2(f)} = f_2\,(1 + m_v\,A_v) = (501.5\ \text{kHz})\,(1 + 1000 \times 0.01) = \mathbf{5.52\ MHz}$$

Note the effect of negative voltage feedback on the bandwidth of the amplifier. The lower cut-off frequency is decreased by a factor $(1 + m_v A_v)$ while upper cut-off frequency is increased by a factor $(1 + m_v A_v)$. In other words, the bandwidth of the amplifier is increased approximately by a factor $(1 + m_v A_v)$.

$$BW_{(f)} \simeq BW\,(1 + m_v\,A_v)$$

where BW = Bandwidth of the amplifier without feedback

BW(f) = Bandwidth of the amplifier with negative feedback